Lab: Design a Network in Yarn

Objective

Design (on paper and optionally in yarn) a crochet piece inspired by a neural network architecture. Choose one of three architectures — feedforward, recurrent, or skip-connection — then draw both the network diagram and the corresponding crochet chart, plan stitch counts and shaping, and (if time permits) crochet the first few rows.

Materials

- Graph paper or blank paper (at least 3 sheets)
- Pencil and eraser
- Colored pencils or markers (at least 4 colors)
- Ruler
- Worsted weight yarn in 2-3 colors (approximately 30 yards total) — for optional crocheting
- Crochet hook, size H/8 (5.0mm)
- Scissors
- Tapestry needle
- Calculator (optional, for stitch count planning)
- Your mesh swatch from the Module 07 lab (for reference)

Architecture Options

Choose ONE of the following three architectures for your design. Read all three before deciding.

Option A: Feedforward Network Scarf (worked in rows) A flat piece worked back and forth in rows, where each row represents a layer of a feedforward neural network. The piece narrows from a wide input layer to a narrow output layer, like a bottleneck network or autoencoder. Stitch types can vary by layer (sc for simple layers, dc for complex layers). No loops, no skip connections — pure feedforward.

Option B: Recurrent Network Hat (worked in the round) A piece worked in the round, where each round represents a time step in a recurrent neural network. The end of each round connects to the beginning, creating the feedback loop that defines recurrence. Plan a hat crown that starts with a small number of stitches (initial hidden state), expands through increases (growing activation), reaches maximum width (full activation), and optionally decreases at the brim (output compression).

Option C: Skip-Connection Textured Panel (worked in rows with post stitches) A flat piece worked in rows that includes front post or back post stitches reaching past the previous row — creating skip connections that allow "information" to bypass intermediate layers. Plan where the skip connections will occur, what stitch they will target, and how they change the texture and structure of the fabric. This mirrors a ResNet architecture with residual connections.

Steps

Part 1: Network Diagram (15 minutes)

1. On the first sheet of paper, draw the neural network diagram for your chosen architecture.

For Option A (Feedforward): - Draw an input layer of 8 nodes (circles) at the bottom. - Draw 2-3 hidden layers, each with progressively fewer nodes (e.g., 8 -> 6 -> 4 -> 2). - Draw an output layer at the top. - Connect each node in one layer to the nodes in the next layer with lines. - Use thick lines for "strong connections" and thin lines for "weak connections."

For Option B (Recurrent): - Draw 6-8 nodes in a circle (representing one round of stitches). - Draw an arrow from the last node back to the first, showing the recurrent connection. - Stack several copies of this circle vertically, connected by arrows, to show multiple rounds (time steps). - Optionally, show the circle getting larger (increases) or smaller (decreases) between rounds.

For Option C (Skip-Connection): - Draw layers as rows of nodes, similar to Option A. - Add bold colored arrows that bypass one or more layers — connecting a node in Layer 1 directly to a node in Layer 3, or Layer 2 to Layer 5. - These skip connections represent the post stitches or spike stitches in your design.

1. Label each layer with the number of nodes. Label the connections with "strong," "weak," or "skip."

Part 2: Crochet Chart Translation (20 minutes)

1. On the second sheet of paper (graph paper is ideal), draw a crochet stitch chart that corresponds to your network diagram. Each square on the graph paper represents one stitch.

For Option A (Feedforward): - Bottom row: the foundation chain, with the number of chains matching the input layer width (e.g., 24 chains for 8 groups of 3). - Each subsequent row: use the stitch chart to show the stitch pattern. Where the network narrows, use decrease stitches (dc2tog, sc2tog) to merge nodes. - Use standard chart symbols: + for sc, T for dc, inverted V for decrease. - Use different colors for different stitch types.

For Option B (Recurrent): - Draw a circular chart (rounds radiating outward from a central ring). - Mark the starting magic ring or chain ring at the center. - Show each round as a concentric ring, with the appropriate number of stitch symbols. - Mark the join (slip stitch or continuous spiral) that creates the recurrent connection. - Show increases as V-shaped symbols where one stitch becomes two.

For Option C (Skip-Connection): - Draw rows of stitch symbols as for a flat piece. - Use bold arrows or colored lines to show where post stitches reach past the previous row. - Mark each skip connection with the target row it reaches to (e.g., "FPdc around dc in Row 2" marked with a colored arrow from Row 4 down to Row 2).

1. Write the stitch counts for each row/round in the margin of your chart.

Part 3: Planning Specifications (15 minutes)

1. On the third sheet of paper, write a planning specification for your piece. Include:

General information: - Architecture type (feedforward / recurrent / skip-connection) - Target dimensions (width x height, or circumference x height) - Yarn weight and hook size - Estimated stitch gauge (stitches and rows per inch)

Stitch plan: - Foundation: number of chains or magic ring size - Row/round by row/round: stitch type, stitch count, increases/decreases, any special stitches - Write at least the first 6 rows/rounds in full pattern notation (e.g., "Row 1: Sc in 2nd ch from hook, sc in each ch across. (23 sc)")

Topology notes: - What is the Euler characteristic of your target surface? (Flat panel: chi = 1. Hat: chi = 2 for a closed top, or chi = 1 for an open tube.) - How do your increases/decreases distribute the curvature budget? - Where are the strong connections? Weak connections? Skip connections?

Network correspondence: - Map each row/round to a network layer. - Identify what the "input" is (foundation) and what the "output" is (final row/round). - Note where the crochet structure most closely mirrors the network architecture and where it diverges.

Part 4: Crochet the Beginning (Optional, 20 minutes)

1. If time permits, crochet the first 4-6 rows/rounds of your design. Use your stitch plan from Part 3.

2. As you crochet, check: does the fabric match your chart? Does the stitch count match your plan? If there are discrepancies, note them — this is prediction error, and resolving it is Active Inference in action.

3. If you chose Option C (skip-connection), try at least one front post or back post stitch that reaches to a previous row. Feel how it pulls the fabric differently than a standard stitch — the skip connection is physically connecting non-adjacent layers.

Part 5: Peer Review and Discussion (10 minutes)

1. Share your network diagram, crochet chart, and planning specification with your crochet circle partner or group.

2. For each design, discuss:

    - Does the crochet chart faithfully represent the network diagram?
    - Are the stitch counts mathematically consistent (do increases and decreases add up)?
    - Could someone crochet this piece from the planning specification alone?
    - What would happen to the "signal flow" if you changed one of the strong connections to a weak one (or vice versa)?

3. If multiple people chose different architectures, compare them. How does the feedforward scarf feel different from the recurrent hat in terms of planning complexity? Which architecture was hardest to translate into crochet?

Discussion Questions

- Was it easier to start with the network diagram and translate to a crochet chart, or to start with a crochet idea and map it onto a network architecture? What does this tell you about the relationship between the two design languages?
- Which neural network concept was most natural to express in crochet? Which was the most difficult? Why?
- How did planning with the Euler characteristic change (or confirm) your intuitions about shaping? Did the math tell you anything you did not already know from experience?
- If you crocheted the first few rows, how closely did the fabric match your plan? What surprised you? What would you change in the next iteration?
- Imagine a crochet circle where everyone designs a different neural network architecture in yarn. What would you learn by holding and comparing the finished pieces?

Wrap-Up

You have completed the final lab of Loop & Lattice. You began this course by discovering that crochet creates topological surfaces. You end it by designing those surfaces intentionally — choosing architectures, planning topologies, translating between the language of neural networks and the language of stitches. The network diagram on your paper and the crochet chart beside it are two representations of the same structure. One lives in the abstract space of computation. The other lives in the physical space of yarn. Both are real. Both are yours.

Keep your designs. Finish your pieces. Bring them to the circle. And remember: every network you crochet is a network you understand — with your hands, with your eyes, and now, with the mathematics that has been hiding in your stitches all along.

Estimated total time: 60 minutes (with optional crocheting: 80 minutes)